A Instructions guide the agent from question to analysis

1 Biological brief

Biological question
Desired measurements
How images were taken

2 Inspect inputs

Channels and scales
Bright / dim regions
Centre and borders

3 Build and run

Choose methods
Set parameters
Run the analysis

Run and inspect the revision

4 Check the result

Raw image and overlay
Same region and scale
Detections and values

5 Repair and recheck

Find the faulty step
Change and rerun
Check other regions too

6 Finalize and deliver

Pipeline and settings
Images and tables
Checks and limitations

The scientist checks images and measurements, and can edit the analysis

B The agent corrects a split nuclear detection (H001)

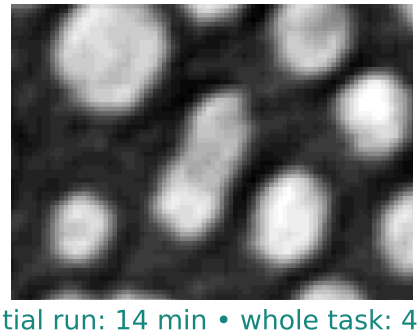
1 • 63 objects
Split body

2 • 60 objects
Split fixed; pair lost

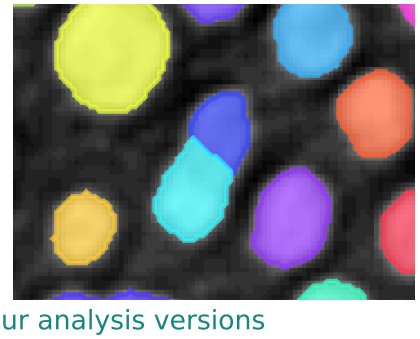
3 • 60 objects
Pair still lost

4 • 61 objects
Pair recovered

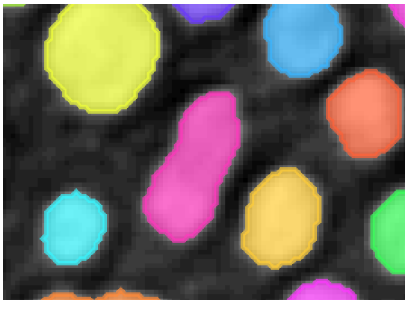
Raw image



First result • split object



Final result • split corrected



Initial run: 14 min • whole task: 45 min • four analysis versions

Counts and decisions: agent's report; whole task includes review, reporting and cleanup.